IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received January 2, 2022, accepted January 5, 2022, date of publication January 7, 2022, date of current version January 19, 2022.

Digital Object Identifier 10.1109/ACCESS.2022.3141311

An Advanced System to Enhance and Optimize Delivery Operations in a Smart Logistics Environment

YASSINE ISSAOUI[1], AZEDDINE KHIAT[1], KARIM HARICHA[1], AYOUB BAHNASSE[2], AND HASSAN OUAJJI[1]

[1]2IACS Laboratory, ENSET of Mohammedia, University Hassan II of Casablanca, Casablanca 20670, Morocco
[2]ISPS2I Laboratory, Ensam Casablanca, University Hassan II of Casablanca, Casablanca 20670, Morocco

Corresponding author: Yassine Issaoui (issaouiyassine818@gmail.com)

ABSTRACT Optimization of order dispatch operations and delivery time prediction is a major concern in supply chains, mainly for e-commerce, which requires the implementation of advanced solutions to reduce delivery time, minimize costs and maximize customer satisfaction. In practice, they fail to warrant scalable and sustainable solutions as the numbers of orders become larger. For that, proper prediction and optimization for delivery operations are required for optimal logistics management. This paper presents an advanced logistics service, which warrants dynamic coordination among all the actors in the smart logistics environment. The proposed advanced shipping system consists of two main parts: the delivery prediction model to compute the expected arrival time, and a hybrid optimization model to tackle path issues. We demonstrate that the advanced system consistently outperforms conventional standard dispatching methods, which means that the proposed approach effectively contributes to optimizing the distribution chain and reducing costs.

INDEX TERMS Smart logistics, deep learning, LSTM, multi-objective optimization, intelligent systems.

I. INTRODUCTION

The global online sales, and e-commerce models development, are showing a speedy growing trend. In 2018, globally, international sales have reached 8.8% of the total online retail expense. From a country point of view, China was the leader in online sales, with the highest records 27.3%($1520.10 billion), followed by the United States 14.0% ($514.84 billion), the United Kingdom 10.9% ($127.98 billion), Japan 4.0% ($110.96 billion), and South Korea 18.1% ($87.60 billion) [1], [2]. Delivery services in e-commerce are mainly based on the logistics process covering different activities i.e.: packaging, storing, distributing, etc. However, these activities require a lot of human and material resources and time as well. Mainly, the low operational efficiency of logistics systems is due to human as well as material factors, especially during the shipping process. For example, since November 11, 2009, a special annual online sales festival is organized by Alibaba, which requires a large number of human and material resources for the dispatching of the placed orders [3].

A set of policies focused on cities, communities, and smart logistics, has been formulated by the European Union (EU) [4]–[6]. These policies request for the further integration of logistics firms into the social responsibilities and smart cities implementation. This requires the adoption of new methods that not only take into account cost, quality, service, time, and environmental requirements but also meet the implementation needs of smart cities. For instance, IBM projected the vision of building an interconnected, advanced, and smart logistics system based on computers, sensors, smart equipment, RFID tags (Radio Frequency Identification), and global position systems [7], [8]. Besides, in China, various technology and logistics players have collectively proposed the concept of smart logistics, to improve analysis, execution, and decision-making capabilities using smart hardware. Although the advanced technologies can be used to increase the automation and intelligence level of the complete logistics system [9]–[11], one of the main challenges that logistics companies must overcome is how smart logistics

The associate editor coordinating the review of this manuscript and approving it for publication was Zhouyang Ren.

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could leverage traditional processes by reducing shipping costs while improving market competitiveness.

In the context of e-commerce, the manufacturing of products sold online takes place in different countries or regions. Consequently, over long distances, logistics companies are principally required to ensure efficient transportation of different types of manufactured products to shipping centers. However, the conducted operations in the shipping centers, at present, are still based on traditional methods to perform the product delivery service. Also, in the shipping centers, to fill up the shipping tasks for the customers having huge product demands, it is required to ensure a short delivery distance and a small average demand. These logistical characteristics not only increase delivery costs but also lower delivery efficiency. Thus, the implementation of smart logistics environments provides efficient and economical solutions for logistics companies. Some important mainstream research works studied smart logistics shipping methods [12]–[15]. Though, few works have discussed the efficiency of intelligent shipping operations using advanced technologies such as Artificial Intelligence (AI), Multi-objective optimization methods, and Internet of Things (IoT).

Smart shipping systems should be able to meet three aspects: the delivery requirements, Secondly, the dynamic cooperation needs, and the current operational process in logistics companies. Associating these aspects, smart logistics systems offer higher efficiency and consistency. In addition, the concern that hampers the smooth running of the logistics process is the shortest path, which reveals three major challenges: i) The pickup path should not collide with other logistics activities; ii) The pickup time should meet the objective of decreasing customer waiting time and enhancing system efficiency; iii) Traditional algorithms should be extended to overcome the weaknesses of the shortest path dilemma. In summary, the design of a smart logistics system must meet the requirements of high efficiency in the context of the fast growth of online sales while ensuring dynamic coordination of all stakeholders.

To better address the above issues, this paper presents an advanced system to enhance and optimize dispatching operations in a smart logistics environment. En effect, our main contributions are summarized as follows:

- An hybrid optimization model to ensure the well-organized and effective coordination of all stakeholders, which involves Dijkstra's algorithm and a hybrid layer, based on Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) algorithms, that supports the delivery operations, especially the dispatching;
- An intelligent dispatching system, based on Deep Learning technique. mainly Uber-LSTM model which allow to predict the delivery time and customer notification.

The rest of this paper is organized as follows: section 2 pictures the studies related to the presented system. Section 3 defines the research problem and the applied methodology in this study; Section 4 describes the proposed advanced system; Section 5 presents the optimization algorithm. Section 6 explains a case analysis, and finally, the conclusion is presented in section 7.

II. LITERATURE REVIEW

This section investigates the smart logistics systems and methods for logistics delivery and highlights the research gaps.

A. SMART LOGISTICS SERVICES

As a key component of the logistics process, traditional delivery services have been the focus of research and industry interests in recent years. Those efforts consist of using various algorithms and methods to solve the route optimization (i.e.: tabu search algorithm, ant colony algorithm, genetic algorithm, etc.) [16], [17]; or to assess and improve the logistics services(i.e.: analytic hierarchy process, fuzzy comprehensive evaluation, etc.) [18], [19]. However, the competitiveness of modern logistics companies is increasing according to emerging technologies, economic variations, and new customer requirements, which has triggered a huge need to evolve logistics services and opt for the concept of smart logistics.

As intelligent life concerns and smart logistics action plans are increasing in different countries, the motivation of logistics companies has extended beyond time, cost, quality, services, automation, and intelligence. Some emerging technologies including AI, IoT, sensor technology, fog, and could technology enhance and strengthen the implementation of smart logistics [20]–[22]. Mainly, the use of advanced technologies increases the logistics system analysis quality, decision-making efficiency, intelligence, and automation level [23]–[25]. Current research works indicate that effective implementation of smart logistics not only improves logistical efficiency and reduces costs [26], but also boosts market competitiveness for logistics companies [27], [28]. For example amazon's automated warehouse [29], [30], JD's unmanned sorting center [31], collaboration and resource sharing [32], and drone delivery [33].

The research on the adoption and implementation of smart logistics services has been extended to different scenarios according to the visions of the research teams, and the economic requirements or needs. For example, Li *et al.* [34] adopted the least square method to resolve the shortest path problems in port smart logistics using cloud technology. Though, in the solution process, this approach is susceptible to fall into local optimal solution, which damages the optimization results. To avoid the local optimal solution, Hu *et al.* [35] presented an improved algorithm based on flower pollination. Although this algorithm ensures meeting the optimization requirements in smart logistics service, in reality, the optimization of smart logistics needs to carefully consider the effects of the e-commerce sector. Accordingly, in an e-commerce environment, Barenji [36] proposed a business logistics model based on an agent synchronization and scheduling method. The principal of this work is to

solve the smart logistics system optimization issues using multi-agent systems. However, wireless network technologies were not studied in this system. Ouyang [9] emphasized that wireless network technology remains essential for smart logistics systems, and it is vital to study their impacts on the traditional logistics models. The smart logistics systems reveal uncertainty factors for instance network congestion. To overcome such issues, Sun [37] proposed an evolutionary mechanism with adaptive parameters, Wang *et al.* [38] proposed an emergency logistics network design problem with the resource sharing under collaborative alliances..

In the logistics dispatching process, customer waiting is a critical issue. Yet, few literature studies have investigated customer waiting time from the perspective of coordination to provide efficient dispatching services, decrease wait times, and promote the market competitiveness of logistics companies. Furthermore, nowadays, logistics managers are pursuing advanced sensible solutions to overcome the issues of insufficient smart operation and low customer satisfaction [39], which forms the main motivation of this research to propose a novel method to solve the stated issues above in logistics dispatching operations.

B. ANALYSIS OF SMART LOGISTICS DELIVERY METHODS

In countries around the world, delivery services rely on various methods that aim to afford effective solutions for solving logistical issues and enhance customers' satisfaction. Beyond providing basic and traditional service, smart logistics aims also to unlock the potentials of logistical operations by enhancing the system abilities and strengthening the effective and equitable delivery to promote economic growth. On this basis, several researchers have made detailed studies on smart logistics methods from a different perspective.

Table 1 confirms that most of the previous methods were focused on the cost, time, quality, service as well as environmental issues in smart logistics delivery. For example, Song [14] adopted a mixed-integer linear programming approach to resolving dispatching time issues. However, this method does not investigate the dynamic factors. Therefore, Kuru [50] solved path optimization in the UAV dispatching process by dynamic multiple allocation technologies in a multi-dimensional space.

Green development has become the key topic of future logistics systems. However, these approaches do not consider the green concept issues, in this setting, the last-mile dispatching issues were studied by Lin (2016) by using the continuous approximation technique. Even if this method considers the green factor, it disregards uncertainty factors. Thus, Lu [54] essentially focused on the solution of practical paths in the plane network without any attention to the possible obstacles. Also, analysis and empirical research methods have appealed to some research scholars. For instance, Otter [43] used the network survey technique to study the preferences of customers in the last mile of dispatching. Also, Perboli [45] studied and evaluated multimodal transport in the last-mile dispatching. Also, Rabe *et al.* [46] proposed a simulation-optimization method to support decision-making for last-mile distribution problems, especially automated parcel locker implementation. However, these surveys did not expose the main steps to reach sustainable dispatching. Therefore, Frehe [44] analyzed the characteristics of group logistics in the dispatching process and revealed the steps to implement sustainable dispatching. Besides, Park [42] investigated courier, express, and parcel (CEP) collaboration in the last-mile dispatching network and proved that the financial viability of CEP cooperation may be warranted only with enormous apartments in the network, to sustain the development of logistics companies.

Algorithm-based methods are another vital issue for smart logistics. For example, based on low cost and low carbon emissions, Xia [49] used a tabu search algorithm to resolve the path optimization problem. Yet, this algorithm falls easily into a locally optimal solution. To tackle this limitation, Tan [52] used an immune cloning algorithm to optimize path problems in dispatching networks, the drawback

TABLE 1. Relevant previous studies.

Methods	Objectives	Ref.
Clarke–Wright saving method-based non-dominated sorting genetic algorithm	smart logistics optimization	[27]
Non-dominated sorting genetic algorithm	Multi-objective path optimization for reverse logistics network	[40]
Logistical network survey	The customers' preferences in the last mile dispatching	[41]
Evaluation and operational research	Customer concerns regarding the quality of dispatching services	[42]
Web survey and performance regression analysis	The last kilometer dispatching services	[43]
Management	Business modeling of urban logistics	[44]
Simulation-optimization	Different dispatching modes integration	[45]
Simulation-optimization	Future automated parcel locker implementations	[46]
The aggregation	The last-mile delivery	[47]
Genetic Algorithm and Extended Particle Swarm Optimization	Secondary logistics dispatching path optimization	[48]
Receding horizon task assignment (RHTA) heuristic	Distribution of unmanned aerial vehicles	[14]
dual objective optimization and Tabu search algorithm	Low-carbon logistics dispatching	[49]
Hybrid agent-based	Exchanging information and making advanced decisions among all stakeholders in smart logistics delivery	[36]
Dimension reduction	Dynamic route optimization	[50]
Operational research and mathematical	Quality and cost of the last mile delivery service	[51]
Improved clone process	Vehicle route problem	[52]
Improved particle swarm optimization	Route optimization for vehicle driving	[53]
The hybrid route planning method	Route optimization for a delivery process	[54]
Reverse labeling Dijkstra algorithm	Path Planning	[55]
Hybrid dispatching	Predictive vehicle dispatching	[56]
Artificial Neural Network (ANN)	Dispatching decision	[57]
Hybrid prediction	Pickup and delivery	[58]

of this algorithm is that it is time-consuming. For this, Li [53] used an improved particle swarm optimization to optimize path dispatching in the cold chain. Similarly, to deal with route optimization issues, Lu [54] adopted the iterative logistics solution planner (ILSP). Zhu *et al.* [55] constructed a novel reverse Dijkstra to solve path planning issues. Chakravorty *et al.* [57] proposed an ANN-based approach to overcome the challenges that influence real-time dispatching decisions. Also, other research teams adopted more neural network-based methods for dispatching and delivery problems [56], [58]. However, these algorithm-based methods still have some limitations to overcome such as the complexity of calculation.

Wang [59] combined both the ant colony algorithm and improved particle swarm optimization to ensure route optimization of secondary dispatching and receiving networks. Frequently, the distribution density of customers affects the dispatching work, but most studies ignore this point. In the case of selecting a dispatching home in dense areas, route issues were solved by Huang [47] using the aggregation method. However, this research lacks the consideration of emergencies, that may cause poor robustness of the logistics system. Therefore, Kokaji [60] studied the case of emergencies when an earthquake occurred, to reach optimal dispatching speed. As observed above, under smart logistics, few pieces of literature focus on optimizing dynamic dispatching. Moreover, various smart technologies have been integrated into the traditional dispatching systems, but these dynamics were not well studied in traditional theoretical methods [20], [61], [62]. Also, in an IoT environment, Wang *et al.* designed an intelligent system to improve dispatching operations [63]. This system highlighted the importance of combining the adopted techniques (Dijkstra and ACO) to reduce the waiting time during the pickup operation, ease package dispatching, and increase customer satisfaction.

Studying the research gaps, it is clear that no method has paid special attention to the coordination between all stakeholders in the logistics environment. However, this could fill the gaps in the system, reduce waiting time, and increase competitiveness. To do this, this method exploits the strengths of smart technologies to consolidate the logistics environment.

C. SMART LOGISTIC IN SMART CITY

Smart cities integrate both digital technology and data analytics to improve the quality and sustainability of life. New technologies, such as smartphones, IA, Blockchain, IoT devices, sensor systems, are directly injected into the lives of residents [64]. Currently, city logistics are undergoing profound changes with the expansion of the urban population and the e-commerce explosion. Smart logistic implementation in smart city can not only provide an efficient delivery service, but also reduce traffic congestion, predict traffic state and so on.

Certainly, each development of the LI will have a positive impact on the implementation of smart cities. Indeed, several works have dealt with this problem, Kuru [65], [66] proposed the concept of STransport integrating intelligent highways, railways, metros, Unmanned Aerial Vehicle Swarms, and air and sea routes for the intelligent management of transport within intelligent cities. Wang [67] proposed a solution for the optimization of logistics networks for fresh products based on a hybrid heuristic algorithm that combines the extended k-means clustering and tabu search nondominated sorting genetic algorithm-II (TS-NSGA-II). Kuru [68] has proposed a framework consists of a decentralized agent-based control architecture that monitors and controls the swarms of resource-constraint FAUAVs for their real-time requirements in optimising their urban uses.

the purpose of this paper is to propose a more general model in the process of intelligent dispatching. Many challenges are still faced, and many factors have not considered. such as, decision-making, smart robot, IoT tools, combination of several DL models, and drone delivery.

III. PROBLEM STATEMENT

The growing technological development has offered advanced solutions to ease people's life such as online shopping [69], [70]. However, the e-commerce managers rely on logistics companies to ensure the orders' delivery, while allowing the customer to choose some delivery options according to the distance, and the waiting time.

Firstly, the main challenge of research works in smart logistics is finding real and significant data to prove the theoretical background. To address this challenge, the paper adopted the data of an important logistics company, which is ''CiaNiao''. As the logistics arm of the well-known Alibaba Group, CiaNiao's principle is to make sure packages get from A-to-B, by relying on Alibaba Cloud's IaaS platform to build a ''logistics cloud'' for packages tracking at every stage of logistics' activities. This company was chosen based on three reasons: i) Practical case study for the research paper objectives, as a professional logistics company, CiaNiao provides various services comprising dispatching; ii) Enhanced competitively, currently, most logistics companies are facing smart technologies and services integration; iii) The Chinese logistics companies, including CiaNiao, are the leaders in that sector, so various researches were studying their services, which is suitable for the results comparison.

In our model, company X refers to ''CiaNiao'', which has a definite dispatching process. Firstly, the ordered package reaches the ''X'' local dispatching center, which is near the customer. Secondly, a short information message is sent to the customer, so that she/he can pick up his package. However, as for other logistics companies, customers have to wait. This can lead to congestion, which decreases the efficiency of the overall logistics activities. Thus, solving this waiting problem will not only ensure the fast accomplishment of the dispatching operation but also improve the operational efficiency of the entire logistics system. To this end, this paper highlights the importance of smart methods and proposes an approach to reduce customer waiting time, improve the efficiency of logistics operations compared to traditional methods, and

establish a solid and scalable foundation for smart logistics environments.

IV. METHODOLOGY

To address the smart dispatching issues, this paper adopts advanced technologies to present a smart dispatching system to enhance smart logistics environments.

A. SYSTEM DESIGN AND PRINCIPALS

Globally, finding the optimal path remains an important problem in logistics systems, which consists of finding an optimal, short, and collision-free path in a given area with various obstacles according to some criterion. For that, the main objective of this system is to support delivery in an e-commerce environment, by proposing a solution for dispatching customer parcels, to optimize time and costs. For that, it is essential to predict the whole delivery time and to locate the shortest path optimally and quickly, while avoiding collisions with other dispatching operations. The detailed smart system is shown in Figure 1.

As there are numerous paths solving techniques, to optimize and enhance the dispatching operations, it is suitable to adopt a graph-based method to ensure better visibility of the logistics activities and avoid possible obstacles. For these reasons, the proposed methodology uses the MAKLINK graph technique for its advantages of speed and calculation volume. However, in smart logistics environments, especially dispatching operations, path optimization is a challenging task, because the system needs to deal with further requirements (e.g. cancelation). The MAKLINK technique is usually used to transform the real physical dispatching network into a free obstacles network [71], [72]. Yet, finding the shortest path remains a laborious task, which needs more than a single method. Consequently, in this paper, to boost the problem solving, we designed a collaborative approach based on several algorithms i.e. Dijkstra, PSO, and ACO.

The specificity of the proposed process is based on a kind of cooperation of models. Firstly, the delivery time prediction model contains different components. The main objective of this model is to maintain decent delivery services. Secondly, the optimization model aims to optimize the time of the dispatching operation. This model is based on MAKLINK graph theory [73] to construct the diagram representing the actual dispatching network. Initially, the environment plan, including the obstacles, is drawn in a 2D coordinate system. In this plane, the vertexes or summits between the obstacles are connected, so the intersection lines with the obstacle are removed. Finally, hybrid optimization is used to recognize the most optimal solutions.

B. DELIVERY PREDICTION MODEL

For forecasting delivery time, multivariate approaches have been widely presented. Although deep learning models, specifically in time series prediction, recurrent neural networks (RNNs) and Long short-term memory(LSTM) have proved promising results [74]–[76]. To the best of our knowledge, the use of deep learning models is still questioned and has not been adopted for advanced delivery services. However, time-series prediction remains a significant and dynamic research area. The main purpose of our work is to improve logistics environments by providing a strong solution. The prediction and optimization remain important for better logistics services and activities. To address these issues, the delivery prediction model aims to provide accurate delivery schedules using a novel deep learning approach to warrant high predictive performance.

In a logistics environment, regardless of the type of goods or services, accurate delivery remains essential to meet customers' expectations as well as delivery requirements. For logistics companies, brand reputation, trust, and relationships are vital to increase future profits and growth and to overcome smoothly the possible issues, that may damage the ongoing activities. For that, several measurements have to be studied to maintain good delivery services. Firstly, calendar vs working days, which is the most common misunderstanding that may occur between suppliers and customers, generally, costumers have this ambiguity; Secondly, the Promise vs real date, suppliers estimate a promise date, which is complicated to meet. The real date needed by the customer may be affected by several circumstances and practices. Thirdly, maintaining accurate inventory records, to make sure that the processes are flawless and robust; fourthly, maintaining realistic schedules regarding the existent capacities; Then, building strong relationships with suppliers to maintain regular reviews on performance. And finally, monitoring logistics efficiency by introducing new means and advanced techniques to enhance the process and meet the customer's requirements.

To predict delivery time more precisely, the proposed model is dependent on timesteps (previous information). For better performance, we combined different components to form our prediction model. Figure 2 (a) presents an overview of the proposed delivery prediction model, which consists of four main parts. Each of these parts performs a task preparing the data for the next task to fortify the prediction process.

1) COLLECTION PART

The principal objective of the goal of the collection part is to compute the weights allowing to inspect the impact of each input variable to seize long-term dependencies. Intuitively, this context-dependent weights generation determines the input to be fed into the following layers of the model. Similarly, to forget gate in LSTM, the collection part network, illustrated in Fig 2-b, determines the input information to forget [77]. Moreover, the use of adaptive weights is significant to perceive the change of some variables over time, especially in a dynamic environment such as le logistics context, to ensure the quick readjustment of input variables [78]. For accurate delivery prediction, additional variables have to be learned by this part i.e.: calendar data, historical variables, special holidays, etc. As the collection part decides which information needs to be considered, it also decides these be forgotten or ignored.

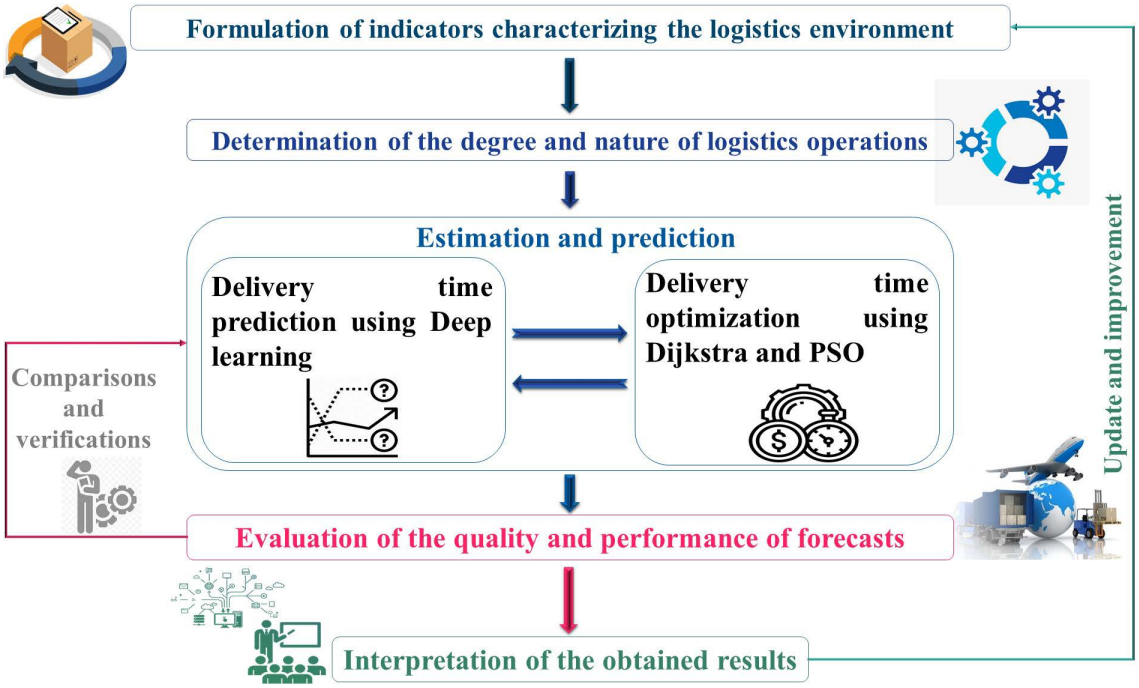


FIGURE 1. A sequence of modeling of the designed smart logistics system.

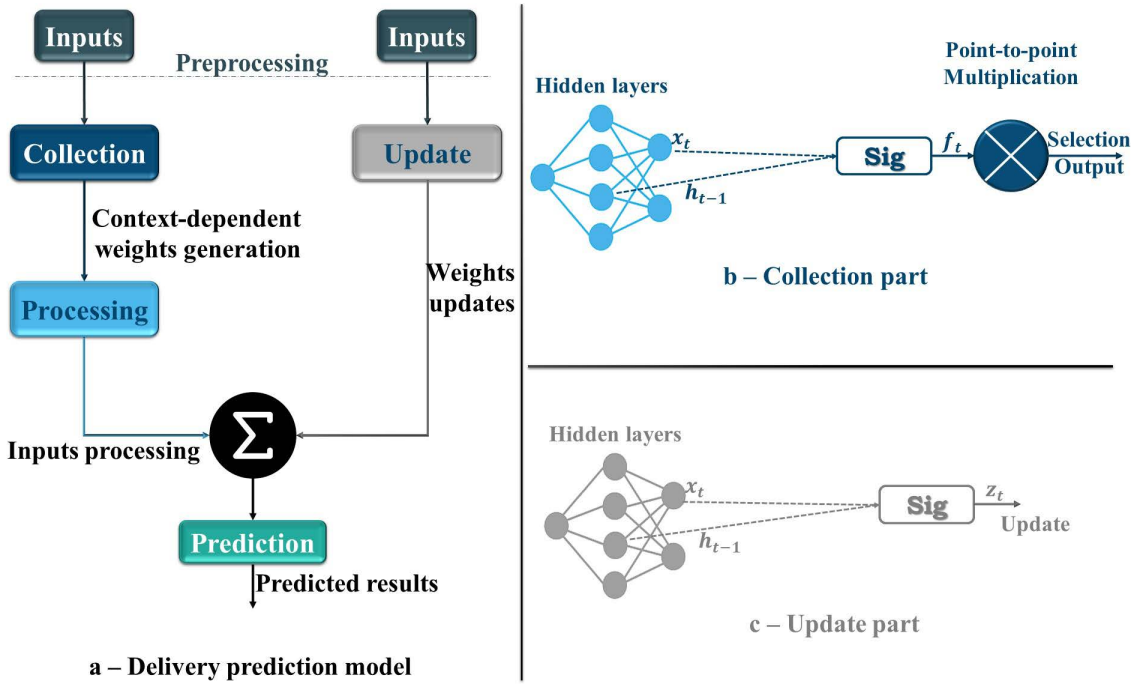


FIGURE 2. The proposed delivery prediction model.

Assuming the use of various n variables such as moving average and calendar data, let $V_t = (v_1, v_2, \ldots, v_T)$ represents a target variable, x_t, h_{t-1} are respectively the input of T timestamps and the previous hidden state, that need to pass through the sigmoid function to produce a value $f(t) \in [0, 1]$ (the forget gate value at time t), and decide whether the information from the old output is still necessary (if the output has a value closer to 1). The value f(t) is calculated as follows:

$$f(t) = sigmoid_{collection}((w_f.[h_{t-1}, x_t] + b_f), \theta) \quad (1)$$

where

- w_f denotes the weight matrix between the forget gate and the input;
- b_f is the connection bias at the time t;

- θ represents the weight parameters for the selected part.

The obtained value f(t) will be used to feed the cell for point-by-point multiplication. Consequently, the generated results denote the context-dependent importance, with values between 0 and 1. For input, a generated weight close to 0 cannot affect the forecasted output. In the delivery prediction model, the collection part decides what logistics information is relevant to keep from the previous environment states, which remains essential to filter the input data for better predictions.

2) UPDATE PART

The proposed Update part is similar to the update gate of the GRU network [79]. This part generates further vectors to update the weighted input vectors, and prepare additional encoded inputs for the prediction part. The proposed update part (Fig 2-c) helps the delivery prediction model to decide how much of the previous information needs to be used in future operations. This allows the model to powerfully copy all the needed information from the historical data while eliminating the risk of gradient vanishment problems.

For a timestep t, the updated value z(t) is calculated using the formula:

$$z(t) = sigmoid_{update}(w^z_{x_t} + u^z_{h_{t-1}}) \tag{2}$$

where $w^z_{x_t}$ denotes the weight of the input x_t, $u^z_{h_{t-1}}$ is weight of h_{t-1}.

Once fed to the network of the update part, the input x_t is multiplied by its weight $w^z_{x_t}$. Similarly, for h_{t-1}, which contains the information of the previous states, it is multiplied by its weight $u^z_{h_{t-1}}$. Then, the results are summed and a sigmoid function is applied to overwrite the result between 0 and 1. Although important information could be generated using the collection part, the additional information from the update part can screen the learned logistics data for better prediction.

3) PROCESSING PART

As neural networks can learn complex relationships, LSTM based models are the most successful networks to directly support multivariate prediction issues. To better analyze the logistics environment, an adequate end-to-end model should be adopted for accurate forecasting. For that, in this part, we used the Uber-LSTM model [80] to handle the obtained sequences from the collection part. Based on prior data, the Uber model contains two main stages: an autoencoder composed of an encoder-decoder; and a fully feed-forward network for trips prediction in a city, or real-time anomalies detection. Furthermore, a skip connection is used to encode the input using the formula:

$$x^{'(t)} = sigmoid_{processing}(x_t) \odot w' \tag{3}$$

where $\odot$ is a wise multiplication, and w' denotes the update weights. One of the difficulties in logistics predictions lies in extreme events forecasting that can damage logistics activities especially delivery i.e.: accidents, special holidays, disasters, emergencies, etc. In response to this issue, we adopted the uber-LSTM model to make the delivery prediction model able to expand single features to multiple dimensions, even with fewer features, to ensure better logistics performance and accurate prediction results.

4) PREDICTION PART

After the previous parts, we typically added the prediction part, which consists of a fully connected layer. This layer takes encoded inputs to forecast the target variable V_t using the formula:

$$V(t) = sigmoid_{prediction}(x'_t, w_t) \tag{4}$$

where w_t denotes the weight parameters of the prediction part. To provide an end-to-end prediction model, the structural prediction part provides an effective process. Firstly, the first layer learns a large number of complex dynamic logistics activities after splicing (the output of selection and update parts). Once the prediction part is implemented, the output weights should be adjusted regarding the obtained results at each time step.

C. OPTIMIZATION MODEL

The proposed optimization model is mainly important to support the coordination within the logistics systems, to solve the delivery issues. The proposed optimization solution relies on the strengths of powerful methods i.e.: Dijkstra, ACO, and PSO. The advantage of Dijkstra's process solves the shortest path problem [81], however, it cannot overcome the logistics path challenges, especially with obstacles. Whereas the advantage of a hybrid optimization is its ability to solve the logistics path issues even in discrete space [53], [59], [71], [82]. Thus, this paper uses these advantages to overcome the difficulty of using ACO or PSO alone with a huge amount of computation. The proposed optimization model details are shown in Fig 3.

The specific optimization process includes two layers. Firstly, Dijkstra's algorithm is used to locate a possible sub-optimal solution in the network space. Then, based on that result a hybrid optimization process is launched to improve the obtained solution. Nevertheless, it adapts two optimization algorithms (PSO and ACO) that compute simultaneously the suboptimal solution. Therefore, it is necessary to evaluate the quality of PSO and ACO solutions. ACO is quite helpful to design and select the resulting features based on the collective ants' rules, which may improve the system performance [83], [84]. PSO is more helpful optimize of the resulting and ancestor features [83]. Consequently, for the best final solution, ACO and PSO have been adopted separately. For solutions ranking, only an optimal solution has been considered using the Simple additive weighting (SAW) method.

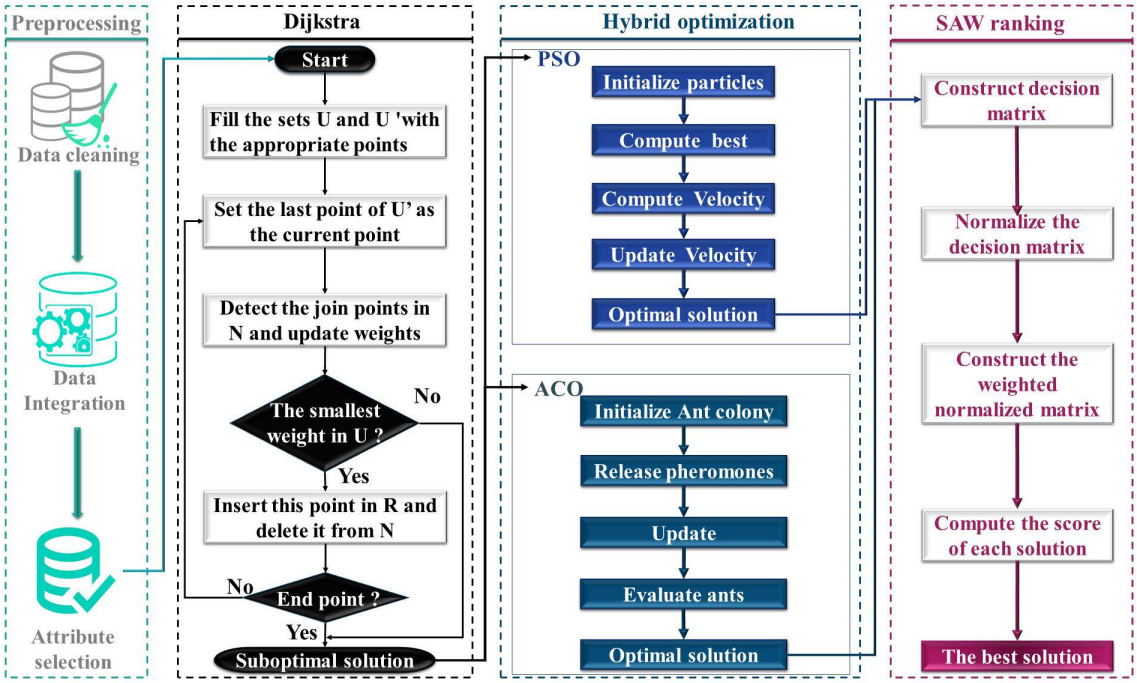


FIGURE 3. The broad architecture of the applied optimization model.

D. DIJKSTRA LAYER

During the dispatching process, finding the shortest path remains an important issue, which may arise due to various practical circumstances. Logistics companies need to optimize their activities especially delivering goods to minimize the waiting time as well as the redundant activities, and to enhance customer satisfaction. In a dispatching problem, within a city, the nodes represent points from which an order can be transported to other points through a specific path. In the graph, an edge is a road that connects two nodes. Practically, the weights of these edges may refer to the length of the path, the cost of transportation, the time, etc.

Figure 4 illustrates an example of the shortest path problem and presents a map of dispatching points. In this weighted graph G, the shortest path should be spotted from o given start point to customer destinations. The example shows that to deliver the 3 parcels there are several possibilities: Customer 1: $0 \rightarrow 1$, Customer 2: $0 \rightarrow 2$, Customer 3: $0 \rightarrow 1 \rightarrow 3$, so following Dijkstra's algorithm the shortest optimal path satisfying all 3 customers is the one marked on the graph on the right, in red, $0 \rightarrow 1 \rightarrow 3 \rightarrow 2$.

To find the shortest path from an initial node to a goal node, using a weighted graph, Lamport proposed Dijkstra's algorithm. This algorithm produces a tree of paths from the initial vertex, to all other nodes in the graph, to find the shortest path [85]. For that, Dijkstra's algorithm is based on the following steps:

- All network's nodes are considered as a whole set S, which consists of these numbered nodes;
- The set $S = P_i \cup P_r$ defines the union of P_i: the set including the initial point, and P_r: the set including all the remaining points;
- The shortest distance D_{sh} between the initial point and its neighbors N_i is calculated, then the nodes spanned by D_{sh} are marked;
- All the marked nodes have recently been added to N_i, then, for the set update, the remaining nodes form the set N_r;
- These steps have to be repeated until the shortest distance Sd between the starting point and the finishing point is reached.

The adopted algorithm to calculate the shortest accessible path is expressed as follows.

After the shortest distance between P_i and P_r is solved, the corresponding shortest path can be marked. The endpoint f belonging to set P_r will be assigned new weight and length values. These steps would be repeated until obtaining the shortest distance from the source node to the end. This result forms the sub-optimal path for dispatching operations.

1) HYBRID OPTIMIZATION LAYER

In path problems, Dijkstra's algorithm is usually prioritized due to its efficiency offer accurate transportation. Dijkstra's algorithm has been proving its capability to spot the shortest path amongst a starting node to completely other nodes. However, this technique loses its performance and reliability when the nodes are mobile. This requires the use of other algorithms that can remedy this context and improve the performance of the logistic environment. On the one hand,

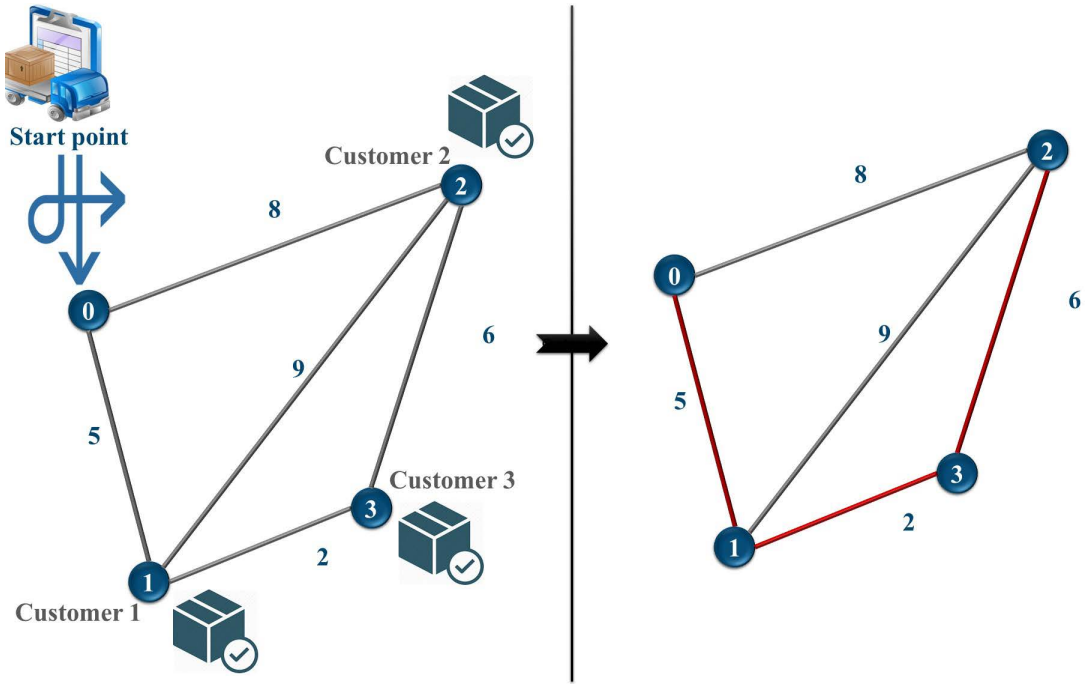


FIGURE 4. Example of dispatching shortest using Dijkstra's algorithm.

ACO and PSO are dedicated techniques for optimization problems in environments with mobile actors. On the other hand, the individual use of one of these algorithms can lead to an increase in execution time or resource consumption [86]. To tackle these issues, this paper introduced a fusion of PSO and ACO techniques.

```
Dijkstra_Shortest_Path_function (G, I)
// G is the graph to compute, I is the initial point
// Initializations
    D_sh [I] = 0                    // Distance from initial node is set to 0
    for each vertex V in G:
        if V ≠ I
            dist[V] = float('inf') // Unknown distances from I to other nodes are set to infinity
        add V to S                  // All nodes initially in S
// The main process
    while S is not empty:
        V = vertex in S with min dist[V] // This vertex is the initial point in the first run-through
        remove V from S
        for each neighbor N of V:    // For the neighbors N still in S.
            dist = D_sh [V] + Length (V, N)
            if alt < D_sh [N] // A shorter path to u has been found
                D_sh [N] = dist          // Update distance of N
    return D_sh []
end function
```

FIGURE 5. Dijkstra shortest path function.

a: ACO ALGORITHM

As an evolutionary heuristic algorithm, Ant colony optimization (ACO) was inspired by the cooperative behavior of real ants. Though the activities of an ant seem very simple, the activities of the whole ant colony are satisfactory. The dynamic of ant society is adopted to solve optimization problems, especially within dynamic environments with increasing activities, numerous individuals, and many information feedbacks. Owing to the advantages of the ACO techniques, it is broadly used to solve several optimization problems, such as goods transportation problems, job-shop scheduling problems, and routing problems [87]–[90]. The ACO algorithm comprises various iterations. In each, the ants use their collected experiences and heuristic information to construct the solutions. The ants' experiences are characterized using the pheromone trail left on the nodes, which constitute a solution.

As ACO is based on the local communication among the artificial ants, their chemical pheromone trails allow them to discover the shortest paths between the nest and the sources. The adopted ACO procedure is described as follows.

The initialization is the first step, in which, many ants are generated randomly. Based on the state of the pheromone, each individual in the ant society has to build a complete solution according to its probabilistic transition rules. Once all individuals generate a solution, the global pheromone is updated. Firstly, a portion of the pheromone evaporates, then each ant produces an amount of pheromone. This second step denotes that the amount of pheromone is intensified after each iteration within an ant search space. Firstly, in the first iteration, each ant has to search randomly for the best solution within the feasible search space (FSS). The feasible search space (FSS) uses the information collected from prior iteration, then is reduced with β to offer more advantages.

Secondly, the amount of pheromone is updated. Finally, based on the old pheromone solution, a new colony is created to compute the optimal solution, which is based on the comparison of the two ants' solutions (old and new). The global optimum solution is investigated utilizing the best values gained from new the colony in the prior iteration.Generally,

```
ACO_function ()
// Initialization
    for i=1 to c // c denotes the number of iterations
        if i=1
            Generate A // A defines the number random ants with FSS
        else
            Decrease FSS with rang [ x_{t-1}^{best} + β ; x_{t-1}^{best} − β ]
        for j=1 to n
            Determine f(x_{t-1}^{best})
            Save x_{t-1}^{best}
        // Pheromone update
        Determine evaporation
        Update pheromone trail
        // Solution step
        Determine search direction
        Generate the values of α vector
        for i=1 to n
            Determine the values of new ant colony
            Determine f(x_{t-1}^{best})_{new} // the new value
            Save x_{t-1}^{best}
            If f(x_{t-1}^{best})_{new} ≤ f(x_{t-1}^{best})
                x_{min}^{global} = (x_{t-1}^{best})_{new}
            else  x_{min}^{global} = x_{t-1}^{best} // the old value
        α_t = α_{t-1} × 0.99
        β_t = β_{t-1} × 0.99
End function
```

FIGURE 6. ACO function.

each ant seeks the social solution while getting maximum profit and optimizing the resources. In a logistics environment, the use of the ACO, only, may cause a problem to reach an optimal agreement within ant society, which can harm the efficiency of the final solution. Such a condition can be tackled by using another optimization technique at the same time, which is particle swarm optimization (PSO).

b: PSO ALGORITHM

This optimization method was inspired by the ordinary movement of birds flying to seek food. This movement established a powerful optimizer, that can be used in several contexts [91]. Like ACO, PSO is based on social behavior. PSO algorithm is based on swarm's behavior to seek the shortest path while having the capacity to detect the optimal options ensuring a mutual agreement of swarms. In this optimization technique, a swarm gathers numerous particles. Each particle as well as its neighbor changes their positions to reach the nodes warranting the spot of the optimal solution. The particles compete to achieve the optimal objectives to avoid local optimality, which can prevent the whole swarm from progressing. To explain better the swarm method let us assume that:

- Each Particle P_i is a part of the swarm S: $P_1, P_2, P_3, \ldots, P_n$;
- The Particle location L_i of a Particle P_i: $L : L_1, L_2, L_3, \ldots, L_n$;
- Each Particle P_i moves within the graph with a velocity or more precisely a speed: $S = S_1, S_2, S_3, \ldots, S_n$;
- To decide the best node to move on, the particle P_i explore the local best location LBL that ensures optimal results: LBL $= lbl_1, lbl_2, lbl_3, \ldots, lbl_n$;
- At a specific time, while moving, across several locations, a particle P_i can explore a global best location GBL: $gbl_1, gbl_2, gbl_3, \ldots, gbl_n$.

```
PSO_function(SHP, P)
// SHP is the shortest path task; P is the set of particles P_i
    Initialize (best) // initialization
    while (SHP≠∅) do // iterate while there still nodes in the path
        for i=0: Np // Np is the number of particles
            B= Evaluation ( P_i ) // using the objective function
            If (B < LBL) LBL =B // update LBL value
                result[i]= LBL
            Update (S, L)
            If (B < GBL) GBL= B
        Sort_Asending (result) // sort the result
        best= min {result [0]}; // select the minimum value
        return (SHP, best) // return the best optimal solution
    End while
End function
```

FIGURE 7. PSO algorithm.

Each particle updates its speed and location giving its individual initial information such as initial speed, the distances from the global best location, etc. These updates occur according to the following process.

As the PSO algorithm search for the optimum solution of the shortest path problem, each particle has different characteristics including the location, speed, and fitness value. The first step aims to prepare the particle by defining input parameters, and constraints. Then, each particle with its location and speed information randomly. The evaluation step seeks the global best location GBL and the local best location LBL, then applies the objective function to assess the fitness values. At this time, the speed and location of particles are updated. Then a comparative step starts, in which, each particle detects its neighboring nodes that ensures the best local solution to compute the LBL.

The PSO technique aims to minimize the computational cost including particles' interactions and shortest path discovery time to offer accurate results. For that, the adopted objective function can be described as follows:

Let I: the set of iterations and SHP_T is the shortest path detection time. The minimization function of I is: $f_1(x) = \sum_{i=1}^{k} I_1$. The minimization function of the path detection time is: $f_1(x) = \sum_{i=1}^{k} T_1$. The formulation of the optimization function could be defined as $Opt_{path}(x) = \sqrt{(f_1(x) - f_{01})^2 + (f_2(x) - f_{02})^2}$ which means that. Where: $T_{i,min} \leq T_i \leq T_{i,min}$, $f_1(x)$, denotes the cost function of I, and $f_2(x)$ is the cost function of the shortest path detection.

The main advantage of the PSO algorithm lies in the spreading of the studied space between several particles, which form swarm' sub-societies. Each particle is focused on an optimal objective of a small part of the problem. This method decreases the complexity level of the shortest path problem. This allows the control of the movement, the speed, and the progress of the path detection.

c: SAW-BASED RANKING

After the processing of both optimization algorithms, the new optimal solutions have been generated. Thus, a prioritizing method is needed to spot the most adequate solution. In the

SAW_function

// Initialization

Construct Decision matrix Dm

//Normalization and construction of the weighted Dm

$$\begin{cases} R_{ij} = \dfrac{Dm_{ij}}{Dm_{ij}^{max}} \\ R_{ij} = \dfrac{Dm_{ij}^{min}}{Dm_{ij}} \end{cases}$$

For i=1 to n //n the number of alternatives

$Pr_i = \sum_{j=1}^{n} wj \times Dm_{ij}$ //Calculate the score of each alternative

Best [i]= Pr_i

Return (max Best [])// The best alternative

End function

FIGURE 8. SAW function.

TABLE 2. Dataset extract.

Package	shelve	X	Y	Priority	Address
A	A1	20	80	5	13 Rue allah ben abdellah Aïn Chock
	A2	20	140	5	Lotissement Al Mostakbal Imm 20 App 13
	A3	20	180	5	Kiniedet Lotissement El Fath 2
B	B1	40	100	5	Bd Omar elkhiam Oasis 23100
	B2	20	120	5	Elriah houbad, Imm 3 App 1
	B3	40	160	5	Imm 5 App 10 Quartier El-hana
C	C1	120	40	5	Ain diab villa 392
	C2	60	40	5	Bd Ghandi Imm 31 App 21
	C3	100	140	5	Bourgone 132 rue asila

context of multi-attribute decision-making, the Simple additive weighting (SAW) procedure remains the most widely used scoring technique [92]. Practically, the SAW technique is mostly used to support systems with overlay operations, especially SIG and logistics environments. To rate the performance of each optimal solution, found by PSO and ACO, SAW relies on the concept of weighted summation to detect the best alternative having the highest score. The algorithm steps are shown below.

Let's consider a set of alternatives $A = A_1, A_2, A_3, \ldots, A_n$. SAW requires the normalization of the decision matrix Dm to a scale similar to the existing alternative ratings. The SAW process is defined as follows:

Once the decision matrix Dm has been constructed, it can be normalized using R_{ij} the normalized performance rating of the alternatives. In the case of maximization, the Dm_{ij}^{max} denotes the maximum value of each row and column. And for the minimization case, Dm_{ij}^{min} the refers to the minimum value of each row and column. Then, the scores of each alternative Pr_i can be calculated where wj is the specified weight. Finally, the best alternative having the maximum score been detected. The advantages of the SAW method lie in its capacity to perform ranking decisions more precisely since it is based on preferences weights and pre-defined values.

V. EXPERIMENTAL ANALYSIS

In this section, the above-presented system is evaluated to analyze its efficiency to optimize important logistics operations, such dispatching process. The implementation framework has been implemented and evaluated using an Intel® Core i7–4700MQ CPU running at 2.40 GHz with 12 GB RAM.

A. DATASET CONTEXT

Recent trends and transitions in the world have increased the choice of online shipping. This sector involves a huge process with several activities. As dispatching is one of the most important pillars of this sector, to assess the proposed system, this paper study the case of a leading logistics company. Company X shoulders a top online shopping market. In this context, it offers packages dispatching services. For that, X is always under pressure to meet the dynamic between customers, pickup agents, logistics obstacles or variations, etc. In some countries, X uses dispatching centers, where customers may look for their packages on site. On the other hand, in other countries, the customer needs to wait for a long period, which may affect the dispatching efficiency. Based on this, the advanced system presented in this paper is used for the dispatch service, so that the models of this paper are preliminarily tested. For this reason, we collected a large dataset from the website of X, after its authorization and following the accesses rules they have assigned to us. This dataset contains 1814892 records with various information. This dataset offers a multitude of order instances, packages, shelves in warehouse, priority, and customer address…, an example of which is shown in Table 2.

Afterwards, we started data pre-processing because the collected data are inconsistent and contain missing values. Data pre-processing enhances the performance because the performance of the deep learning algorithms depends on the quality of input data. The raw data collected from the X Company contain many missing and erroneous values, which occur due to the usage of faulty measuring instruments, incorrect data entry. Additionally, when the data are scattered over a large scale, it makes the analysis difficult and increases the execution time.

B. SYSTEM PERFORMANCE

As delivery and dispatching services are quite linked, the implemented system proposes a delivery prediction system to well organize the dispatching tasks, which in turn are optimized using the optimization model. To evaluate the performance of the presented system, a set of tests has been established. First, it is essential to test the prediction model to see how accurate the predicted delivery times are. This set of tests will evaluate the performance of each component separately, and then compare it with the overall performance of the prediction model. Secondly, the dispatching optimization test set will take into account both layers and their fundamental algorithms. Finally, a comparative study would be done regarding relevant literature works.

1) THE PREDICTION MODEL TESTS

As Neural networks form the baselines of our prediction model, a set of hyperparameters were set up for computation time optimization. The model was trained using an activation function = linear, sigmoid, tangh, a batch size of 8,16, and a learning rate of 0.01,0.001.

To evaluate the model's performance, three main evaluation metrics were adopted i.e.: mean absolute error (MAE), the mean absolute percentage error (MAPE), and symmetric mean absolute percentage error (sMAPE):

$$\begin{cases} MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \\ MAPE = \frac{1}{n}\sum_{i=1}^{n}|\frac{(y_i - \hat{y}_i)}{y_i}| \\ sMAPE = \frac{1}{n}\sum_{i=1}^{n}|\frac{(y_i - \hat{y}_i)}{(\hat{y}_i + y_i)}| \end{cases}$$

where n denotes the number of test series datasets, $\hat{y}_i$ symbolizes the ith forecasted value, and y_i represents the ith current value. The more the MAPE value is smaller than 0.15, the best is the efficiency of the model is.

As the prediction model is a set of collaborative parts, we decided to test the performance of each component and compare it with the results of the proposed collaborative model. Tables 3, 4, and 5 show the prediction performances of the proposed models for each dispatching center. The prediction results show that our model, as a collaboration of deep learning models, outperforms other components models on most of the dispatching centers' data.

The collection and processing parts did not achieve good performances, by working individually, on most datasets. This fact can be due to data preprocessing because while preparing the input datasets, only the historical data were used for each time step, and the time step of the model's part was set by 1.

Though time series data were prepared and performed long-term prediction for the UBER-LSTM and LSTM based parts, the use of the predicted value may increase the errors. Thus, to enhance the predictive performance, we proposed a novel deep learning model based on combined neural networks architectures consisting of collection and update layers, which are similar to gate mechanisms in LSTM and GRU networks. The proposed prediction model outperformed the prediction results of the individual parts.

Exploring these results, it is obvious that the prediction model can be used to estimate the volume of delivery packages available even on holidays. Furthermore, this model can help to perform e-commerce delivery by efficiently computing the required costs for delivery.

2) THE OPTIMIZATION MODEL TESTS

For the optimization experimentations, the operational parameter settings are shown in Table 6. Only the best tuning parameters have been adopted in the case of both ACO and PSO algorithms, to reach optimal results and enhance dispatching accuracy.

To evaluate the optimization model's performance, we adopted two main metrics i.e.: the average error rate (AER), and the accuracy (ACC).

For the AER computation, the mean percentage error has been calculated as the average ratio errors by the variance from the model and the actual values. Figure 5 shows the average AER, and Accuracy comparison in the case of PSO and ACO functioning. The exposed results show that the error rate is minimal in the case of SAW comparing to PSO or ACO.

The optimization model evaluation demonstrates the importance of applying both ACO and PSO algorithms in case of shortest path issues and dispatching in logistics along with the simple additive weighting comparison technique. The obtained results are unbiased as the solution is completely optimized. The hybrid optimization model has proved high-performance rates with promising possibilities to ensure the sustainability of the logistics environments.

As either ACO or PSO alone show modest results, their combination can lead to a more robust and promising result. This combination along with the SAW method is much promising result to enhance the dispatching services.

3) RESULT ANALYSIS

This comparative study aims to reveal the potential of the advanced system presented in this work in comparison with previous relevant literature works. We first study the results of the proposed method. Using the same datasets, we further investigate the performance of leading previous works i.e.: a- Yong *et al.* [44], b- Lu *et al.* [50], c- Zhu *et al.* [51], d- Wan *et al.* [52], e- Chakravorty *et al.* [53].

To assess the speed of system prediction, the variations of average values are shown in Figure 6. The hybrid optimization layer has more advantages than the individual ACO or PSO optimization algorithms, which is principally reflected in two aspects. Firstly, the hybrid optimization has a faster convergence speed, which remains vital to improve the ability of the dispatching system to respond to time requirements. Secondly, the hybrid optimal value (30,8) is much better than the average value obtained by another based optimization algorithm.

To further investigate the effect of the hybrid optimization layer on the performance, we evaluated the average lead time of the proposed system, and the incomplete system (without hybrid optimization). The experiment results are displayed in Figure 7. When the execution time is between [0–100] s, both the average lead time values deteriorate as the time increases. Yet, this decline becomes slower. At this range, incomplete system and the whole system nearly have the same performance due to the small number of tasks. In this case, both systems can identically perform delivery services well. However, as the processing time continues increasing, more logistics activities will be taken as the number of new orders exceeds the number of available orders. Then the incomplete system without any optimization remedy falls into the issue that many orders may go to dispatching centers and

TABLE 3. MAE of the prediction model results on dispatching centers'.

Dispatching centers	Collection part	Update part	Processing part	Prediction part	Whole model
Dispatching center - 1	97864,19	111362,39	172104,39	44508,99	33747,99
Dispatching center - 2	16677,09	17371,79	696,89	33534,09	9729,19
Dispatching center - 3	8122,69	6248,69	5311,69	4224,09	3750,99
Dispatching center - 4	4401,59	26399,09	29332,09	25839,49	23466,09
Dispatching center - 5	46183,59	32189,19	64376,29	20428,09	6999,29
Dispatching center - 6	1258,79	420,29	22168,79	4114,79	5857,39
Dispatching center - 7	26326,39	17748,79	26917,89	8047,49	9171,19
Dispatching center - 8	62215,39	52644,09	67000,99	21641,59	17549,39
Dispatching center - 9	15820,69	11865,99	5538,59	39769,79	9493,19
Dispatching center - 10	39398,29	33604,79	40557,09	19389,49	17382,79
Dispatching center - 11	58182,29	40885,49	48747,59	37310,89	7864,29
Dispatching center - 12	36003,89	29192,69	42814,99	18431,79	14597,39
Dispatching center - 13	15191,59	17564,99	15666,29	16644,19	11968,89
Dispatching center - 14	11648,99	5624,69	8837,69	6845,39	6829,59
Dispatching center - 15	31445,99	33351,59	35257,29	15129,19	10483,39
Dispatching center - 16	26521,79	9412,29	14545,19	9125,99	11978,69
Dispatching center - 17	6335,79	2852,19	2218,89	3552,99	4118,99
Dispatching center - 18	10070,99	421,59	6714,69	6010,29	5456,09
Dispatching center - 19	16684,49	25581,79	27806,09	20511,49	11123,69
Dispatching center - 20	13538,39	4121,19	15301,79	20669,79	8828,79
Average	**27194,645**	**23943,18**	**32595,76**	**18786,495**	**11519,865**

TABLE 4. MAPE of the prediction model results on dispatching centers'.

Dispatching centers	Collection part	Update part	Processing part	Prediction part	Whole model
Dispatching center - 1	0,3	0,34	0,52	0,14	0,11
Dispatching center - 2	0,25	0,26	0,02	0,49	0,15
Dispatching center - 3	0,27	0,21	0,18	0,15	0,13
Dispatching center - 4	0,04	0,19	0,21	0,19	0,14
Dispatching center - 5	0,34	0,24	0,47	0,16	0,06
Dispatching center - 6	0,04	0,02	0,54	0,16	0,15
Dispatching center - 7	0,9	0,61	0,92	0,28	0,22
Dispatching center - 8	0,4	0,34	0,43	0,15	0,12
Dispatching center - 9	0,21	0,16	0,08	0,51	0,13
Dispatching center - 10	0,35	0,3	0,36	0,18	0,16
Dispatching center - 11	0,38	0,27	0,32	0,25	0,06
Dispatching center - 12	0,38	0,31	0,45	0,2	0,16
Dispatching center - 13	0,33	0,38	0,34	0,36	0,2
Dispatching center - 14	0,3	0,15	0,23	0,18	0,16
Dispatching center - 15	0,34	0,36	0,38	0,17	0,12
Dispatching center - 16	0,63	0,23	0,35	0,22	0,21
Dispatching center - 17	0,21	0,1	0,08	0,12	0,14
Dispatching center - 18	0,25	0,02	0,17	0,15	0,14
Dispatching center - 19	0,16	0,24	0,26	0,19	0,11
Dispatching center - 20	0,24	0,08	0,27	0,36	0,16
Average	**0,316**	**0,2405**	**0,329**	**0,2305**	**0,1415**

wait for a long time instead of forwarding the already arrived orders.

In practice, the order's pre-arrival information may contain forecasting errors. For that, we considered four evaluation metrics i.e.: the mean average absolute error (MAE), the mean absolute percentage error (MAPE), the symmetric mean absolute percentage error (sMAPE), and the R2 score.

Table 6 provides an overview of the forecasting system. The furthermost remarkable result is that the advanced proved its strength in terms of accuracy with a MAPE of 0,122. Generally, a MAPE value lesser than 0.15 is considered a successful system with major enhancement. As seen in Table 6, when delivering orders, our optimization-based method converges to metrics values indication a solution close to the optimum, even with more orders. Besides, the coefficient of determination R2 proved important results near to 1 even with more orders, which means that the proposed system has the best possible scores.

To verify the effectiveness of our advanced system, we further compare its performance with prior works. Table 8 illustrates the average results of evaluation metrics of the tested methods. It is shown that these methods are all competitive with good performance rates. Nevertheless, it is obvious that the proposed method outperformed the other relevant literature method by providing optimal results, which can highlight the importance of the proposed system in logistics environments.

In other words, the proposed intelligent advanced system is promising to enhance the logistics systems. The cooperation between the deep learning and optimization models is an

TABLE 5. sMAPE of the prediction model results on dispatching centers'.

Dispatching centers	Collection part	Update part	Processing part	Prediction part	Whole model
Dispatching center - 1	0,38	0,2	0,25	0,15	0,06
Dispatching center - 2	0,94	0,15	0,15	0,29	0,11
Dispatching center - 3	0,16	0,15	0,15	0,11	0,04
Dispatching center - 4	0,33	0,22	0,28	0,24	0,15
Dispatching center - 5	0,77	0,19	0,42	0,27	0,08
Dispatching center - 6	0,33	0,22	0,26	0,2	0,08
Dispatching center - 7	0,73	0,47	0,71	0,2	0,23
Dispatching center - 8	0,37	0,21	0,27	0,12	0,08
Dispatching center - 9	0,26	0,18	0,34	0,22	0,1
Dispatching center - 10	0,3	0,24	0,29	0,22	0,14
Dispatching center - 11	0,31	0,22	0,4	0,21	0,14
Dispatching center - 12	0,34	0,21	0,27	0,25	0,14
Dispatching center - 13	0,23	0,49	0,36	0,17	0,12
Dispatching center - 14	0,5	0,2	0,22	0,19	0,12
Dispatching center - 15	0,25	0,25	0,25	0,22	0,04
Dispatching center - 16	0,48	0,22	0,3	0,43	0,77
Dispatching center - 17	0,24	0,16	0,18	0,16	0,04
Dispatching center - 18	0,2	0,2	0,16	0,15	0,05
Dispatching center - 19	0,22	0,11	0,13	0,31	0,04
Dispatching center - 20	0,32	0,41	0,3	0,4	0,23
Average	**0,383**	**0,235**	**0,2845**	**0,2255**	**0,138**

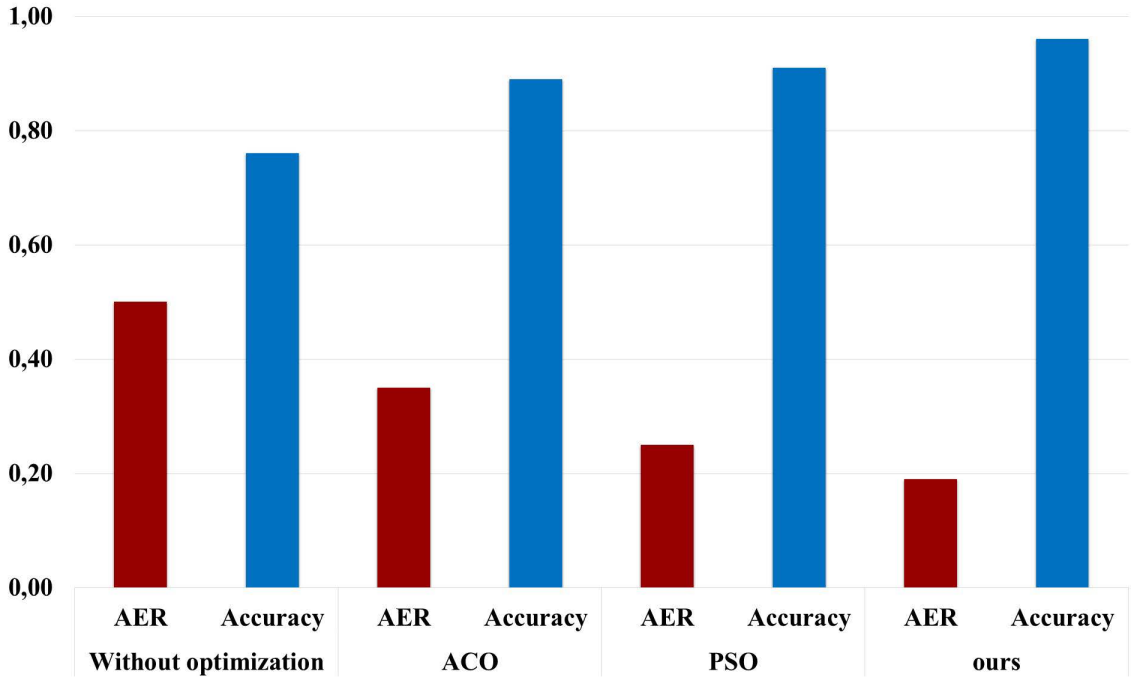


FIGURE 9. AER and accuracy comparisons.

TABLE 6. Parameter settings for the experiment runs.

ACO algorithm	PSO algorithm
Initial number of ants: 50	Initial number of particles :50
α=1, β =1, C=10	ω=0.6289
Evaporation factor ρ = 0.4	c_1=1.71, c_2=1.98,
Pheromone Trail: φ = 0.5	T_{max} = 15 sec

improved mode; that contributes to making dispatching services more dynamic and prominent. The system computes the optimal arrival time, so that customers may receive their order with the minimal waiting time. This system can be applied to minimize the delivery time and increase customers' satisfaction. Theoretically, the minimization of the waiting time contributes to improving the overall efficiency of logistics services pickup. Furthermore, to reduce the dispatching costs of the enterprises, this paper adopted the shortest path as an optimization. The use of various algorithms and technologies is beneficial to enhance the traditional dispatching systems as well as the whole logistics environment.

The proposed solution could be very promising in terms of management insights, specifically in:

- Minimization of delivery costs;
- Reduction of delivery times;

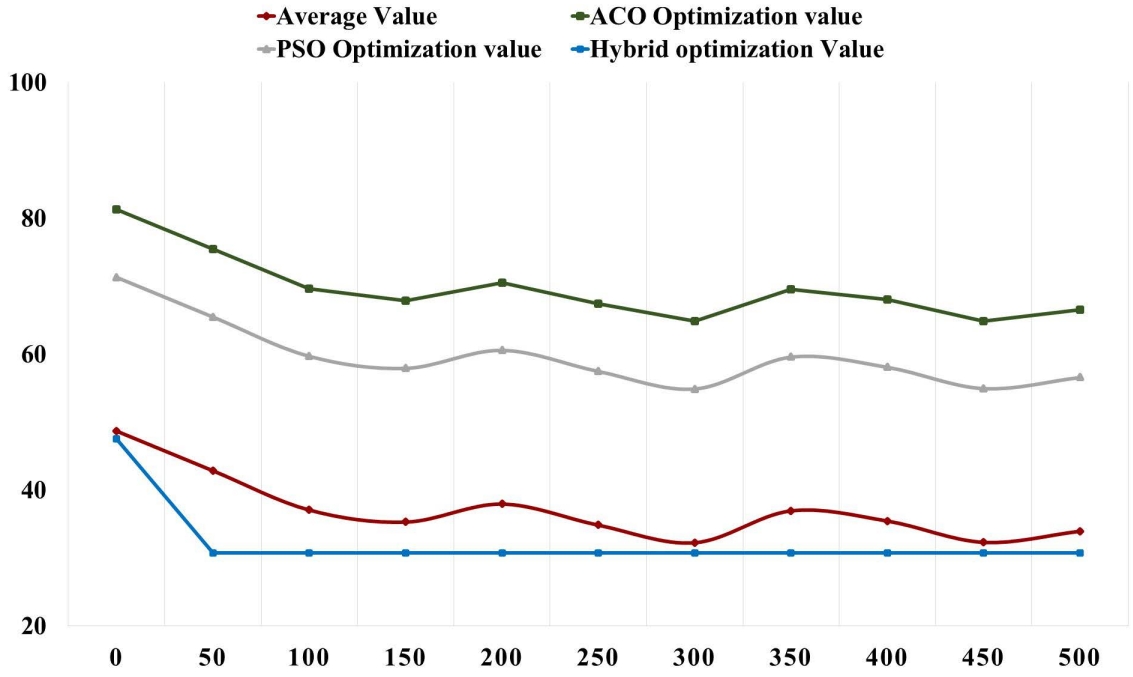


FIGURE 10. The evolution of system average values of different algorithms.

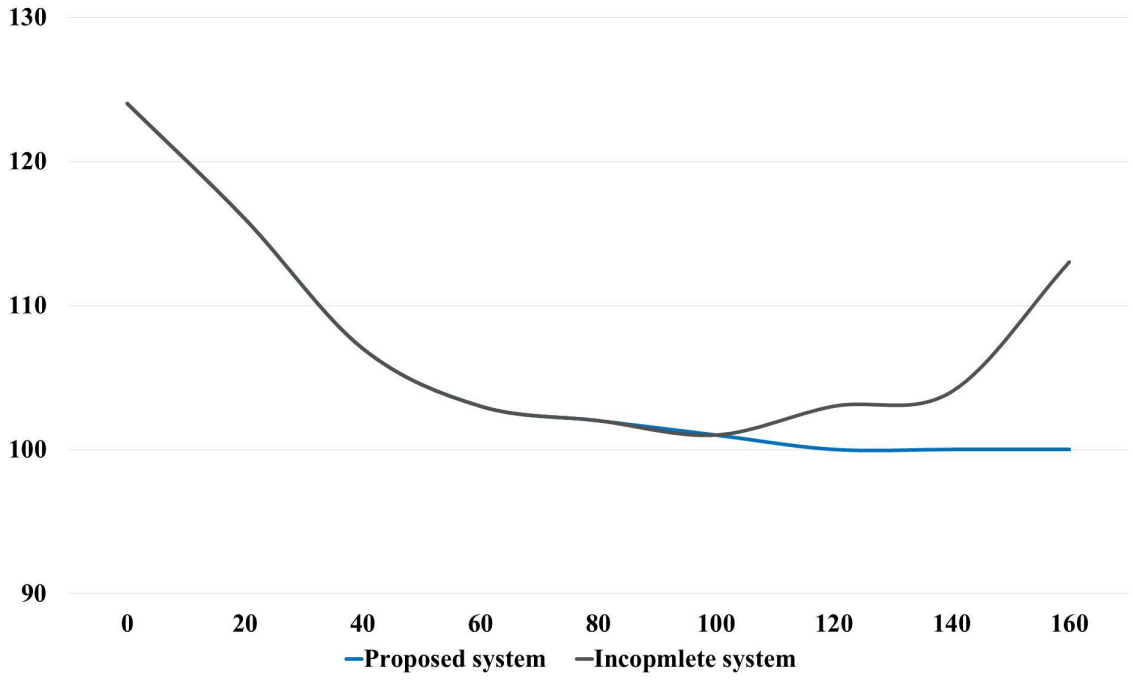


FIGURE 11. Performance of the proposed system.

- Improvement of the notifications, alerts and information system;
- Maximization of customer satisfaction;
- Increasing the competitive level;
- The increase in commercial margins;
- The development of the activity.

VI. CONCLUSION

Our study proposed an advanced dispatching system in a smart logistic environment, based on a hybrid approach composed of artificial intelligence techniques and optimization algorithms. In reality, this study has a significant contribution based on three aspects. Firstly, the paper presents a detailed

TABLE 7. Overview of the forecasting system regarding the orders' number.

Orders	MAE	MAPE	sMAPE	R^2 score
10	33745,9	0,1	0,07	0,955
20	9727,1	0,14	0,1	0,974
30	3748,9	0,12	0,11	0,967
40	23464	0,13	0,13	0,955
50	6997,2	0,05	0,12	0,964
60	5855,3	0,12	0,13	0,977
70	9169,1	0,14	0,13	0,982
80	17547,3	0,11	0,14	0,976
90	9491,1	0,12	0,14	0,981
100	17380,7	0,15	0,15	0,988
Average	**13712,66**	**0,118**	**0,122**	**0,9719**

TABLE 8. Performance' comparison.

	MAE	MAPE	sMAPE	R^2 score
a - Yong et al.	62625,92	0,141	0,141	0,8465
b - Lu et al.	75496,92	0,143	0,142	0,85937
c - Zhu et al.	59606,77	0,138	0,36	0,89511
e - Chakravorty et al.	27826,47	0,122	0,127	0,96798
f- Our method	**13712,66**	**0,118**	**0,122**	**0,9719**

idea about designing a novel system, including how intelligent technologies are applied to ease the dispatching and picking up of packages. Secondly, this work describes the dispatching operation, including the delivery prediction time, and the optimized path planning. Thirdly, this research is verified by adopting a case study of a global leading logistics company, which is helpful for managers to better carry forward in practice.

Theoretically, the results of this study also highlight important contributions. Because, at present, there are rather few research results on the cooperative performance of logistics environment, customer satisfaction, and intelligent technologies, so, in this work, a novel method was designed to solve this issue, which can fill the research gaps and lacks in the smart logistics field. Specifically, in the era of deep learning enhancement, implementing advanced dispatching has become a new tendency. In contrast, this paper describes a novel delivery prediction model as well as a strong hybrid path optimization model. In addition, the superiority of this approach is also demonstrated by the performance results. Of course, the coordination of all stakeholders is also an important factor, which is frequently ignored in previous literature.

Computational Results clearly show that our proposed method outperforms ACO and PSO methods. Firstly, the hybrid optimization has a faster convergence speed, which remains vital to improve the ability of the dispatching system to respond to time requirements. Secondly, the hybrid optimal value is much better than the average value obtained by the others based optimization algorithms.

The limitation, of this work, is the evaluation of the designed method in this paper using collected old case data instead of real-time case data. Nevertheless, this does not impact the theoretical and practical worth of this study; the purpose of this paper is to make the novel models more general. Furthermore, in the process of intelligent dispatching, many challenges are still faced, such as decision-making, smart robot, IoT tools, etc. However, this work did not consider those factors. In future works, we will focus on solving the issues of smart robot adoption, failures, and improvements in logistics systems.

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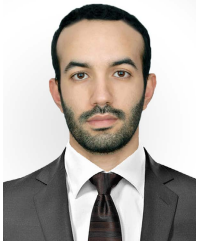
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YASSINE ISSAOUI received the M.S. degree from the Hassan II University of Casablanca, in 2014, where he is currently pursuing the Ph.D. degree with the Signals, Distributed Systems and Artificial Intelligence Laboratory. His research interests include artificial intelligence, machine learning, deep learning, blockchain, project management, smart logistic, the Internet of Things, industry 4.0, smart supply chain management, smart manufacturing, smart grid, and smart city. He is an outstanding reviewer of various indexed journals. He has certified on several products and solutions.



AZEDDINE KHIAT received the H.D.R. degree in computer science and Ph.D. degree in computer science, networks, and telecommunications from the ENSET Mohammedia, University Hassan II of Casablanca, Morocco. He is currently a Professor and a Researcher with the Department Mathematics and Computer Science, ENSET Mohammedia, University Hassan II of Casablanca. He is also a Research Member with the Computing, Artificial Intelligence and Cyber Security Laboratory (2IACS), Artificial Intelligence, Cybersecurity and Digital Trust Team (IACSCN), ENSET Mohammedia, University Hassan II of Casablanca. He has published tens of refereed journal and conference papers. His research interests include wireless networks, QoS on mobile networks, handover on wireless networks, networks, and telecommunications, SDN, new generation networks, security, mobile learning, the IoT, smart city, smart grid, and industry 4.0. He has certified on several products and solutions. He is an organizing committee member and a technical program committee member of tens of international conferences. He is an outstanding reviewer on various indexed journals.



KARIM HARICHA received the M.S. degree from the Hassan II University of Casablanca, in 2014, where he is currently pursuing the Ph.D. degree with the Signals, Distributed Systems and Artificial Intelligence Laboratory. His research interests include artificial intelligence, machine learning, deep learning, smart manufacturing, the Internet of Things, industry 4.0, smart supply chain management, smart logistics, smart grid, and smart city.



AYOUB BAHNASSE received the Ph.D. degree in networks and telecommunications from Chouaib Doukkali University, Morocco. He is currently a Professor and a Researcher with ENSAM Casablanca, Hassan II University of Casablanca, Morocco, where he is also a Research Associate with the Information Treatment Laboratory, Faculty of Sciences Ben M'sik. He is a member of "Big data and Internet of Things project for urban service" Project, Hassan II University of Casablanca. He has published tens of refereed journal and conference papers. His research interests include new generation networks, security, mobile learning, wireless networks, quality of service, multimedia systems, the IoT, smart city, and MPLS. He is an organizing committee member and a technical program committee member of tens of international conferences around 20 countries. He is an Outstanding Reviewer of various indexed journals, such as *Computer Networks* journal (Elsevier), *Computers & Electrical Engineering* (Elsevier), *Computers & Security* (Elsevier), IEEE TRANSACTIONS ON COMMUNICATIONS, *Telecommunication Systems* (Springer), *Wireless Personal Communications* (Springer), *Mobile Networks and Applications* (Springer), and tens of indexed journals. He has certified on several products and solutions among them: Fortinet NSE4, Microsoft Azure Administrator Associate, Juniper DevOps Associate, Juniper Cloud Associate, Juniper Design Associate, and Huawei RS Associate.



HASSAN OUAJJI received the degree (Hons.) from the Department of Electronic and Electrical Engineering, ENSET Mohammedia. He is currently a Full Professor with the Department of Electronic and Electrical Engineering, ENSET Mohammedia. He is also a Research Member with the SSDIA Laboratory and the Head of the Network and Telecommunications Team, Hassan II University of Casablanca, Morocco. He has published tens of refereed journal and conference papers. He is a member of several scientific project with the Hassan II University of Casablanca. His research interests include acoustic electrical engineering, signal processing, electronic, networks, and telecommunications. He is an organizing committee member and a technical program committee member of tens of international conferences. He is an outstanding reviewer of various indexed journals.